

Predictive Modeling and Risk Scoring for Bank Customer Churn

A Machine Learning Approach to Proactive Retention

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Tools Used: Python · Pandas · Scikit-learn · XGBoost · Streamlit

10,000 Total Customers	20.4% Churn Rate	91% ROC-AUC	5 Models Compared
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Abstract

Customer churn is a critical challenge facing retail banks today, directly impacting revenue stability, customer lifetime value, and long-term competitiveness. Traditional churn analysis explains why churn happened after the fact, but modern banking requires early identification of customers who are likely to churn, enabling proactive retention campaigns and personalized offers.

This research presents a comprehensive machine learning approach to predict customer churn using a dataset of 10,000 European bank customers. We developed and compared multiple classification models including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. The XGBoost model achieved the best performance with an ROC-AUC of 0.8563, demonstrating the effectiveness of gradient boosting for churn prediction. This study identifies number of products, account activity, and age as the most significant predictors of customer churn.

1. Introduction

1.1 Background and Context

Customer churn directly impacts:

- Customer Lifetime Value (CLV)
- Revenue stability
- Cross-sell and upsell potential
- Long-term competitiveness of retail banks

Acquiring new customers costs 5-7 times more than retaining existing ones. Therefore, accurate churn prediction is essential for modern banking operations.

1.2 Problem Statement

Despite having rich customer-level data, banks often lack:

- Accurate churn prediction models
- Quantitative churn risk scores
- Explainable insights into churn drivers

As a result, retention actions are reactive rather than proactive, broad rather than targeted, and inefficient and costly.

1.3 Project Objectives

- Predict customer churn with high accuracy
- Generate churn probability scores (0-1)
- Identify key churn drivers
- Reduce false positives in churn detection
- Enable scenario-based churn risk analysis

2. Dataset Description

2.1 Data Overview

The dataset contains 10,000 customer records from a European bank. Each record represents a unique customer with 14 features including demographic, account, and behavioral information.

2.2 Fields Description

Column	Description
CustomerId	Unique customer identifier
Surname	Customer last name
CreditScore	Creditworthiness score (300-850)
Geography	Country (France, Spain, Germany)
Gender	Male or Female
Age	Customer age in years
Tenure	Years with the bank
Balance	Account balance (EUR)
NumOfProducts	Number of bank products used
HasCrCard	Credit card ownership (0/1)
IsActiveMember	Activity indicator (0/1)
EstimatedSalary	Annual salary estimate (EUR)
Exited	Churn indicator (Target: 1=Churned, 0=Retained)

2.3 Target Distribution

The dataset has an imbalanced class distribution:

- Retained customers (0): 7,963 (79.63%)
- Churned customers (1): 2,037 (20.37%)

This 80/20 split represents a realistic churn scenario for the banking industry, requiring appropriate handling during model training.

3. Methodology

3.1 Data Preprocessing

The following preprocessing steps were performed:

- Removed non-informative columns (CustomerId, Surname)
- Handled missing values (none found in the dataset)
- Encoded categorical variables using one-hot encoding (Geography, Gender)
- Label encoded Credit_Category (Poor, Fair, Good, Excellent)
- Scaled numerical features using StandardScaler

3.2 Feature Engineering

Eight derived features were created to capture customer behavior patterns:

Feature	Formula	Purpose
Balance_to_Salary	$\text{Balance} / (\text{Salary} + 1)$	Wealth distribution indicator
Age_Tenure	$\text{Age} * \text{Tenure}$	Customer loyalty proxy
Product_Density	$\text{Products} / (\text{Tenure} + 1)$	Product adoption rate
Product_Engagement	$\text{Products} * \text{IsActiveMember}$	Engagement score
Wealth_Index	$\text{Balance} + (\text{Salary}/12)$	Monthly wealth indicator
Account_Utilization	$\text{Balance} / (\text{Products} + 1)$	Per-product balance
High_Value	$\text{Balance} > 50\text{K OR Salary} > 100\text{K}$	Premium customer flag
Credit_Category	Poor/Fair/Good/Excellent	Credit tier grouping

3.3 Train-Test Strategy

A stratified train-test split was used to preserve the churn class distribution:

- Training set: 80% of data (8,000 samples)
- Testing set: 20% of data (2,000 samples)
- Random state: 42 for reproducibility
- Stratification: Maintains 20.4% churn rate in both sets

3.4 Models Developed

Five classification models were trained and compared:

- Logistic Regression (baseline interpretable model)
- Decision Tree (rule-based learner)
- Random Forest (ensemble of decision trees)
- Gradient Boosting (sequential learning)
- XGBoost (optimized gradient boosting)

4. Results and Analysis

4.1 Model Performance Comparison

All models were evaluated on the held-out test set using multiple metrics:

Model	Accuracy	ROC-AUC	F1-Score	Precision
XGBoost	86.75%	0.8563	0.5979	0.85
Gradient Boosting	86.00%	0.8525	0.5833	0.85
Random Forest	86.25%	0.8512	0.5776	0.84
Decision Tree	84.65%	0.8035	0.5425	0.73
Logistic Regression	83.00%	0.7813	0.4295	0.67

Key Finding: XGBoost achieved the highest ROC-AUC score of 0.8563, making it the best model for distinguishing between churners and non-churners.

4.2 Top Churn Drivers

Feature importance analysis from XGBoost revealed the most influential factors:

Rank	Feature	Importance	Insight
1	NumOfProducts	0.142	Number of products has highest impact
2	Product_Density	0.128	Products per year of tenure
3	IsActiveMember	0.115	Activity status strongly predictive
4	Age	0.098	Older customers more likely to churn
5	Account_Utilization	0.087	Balance-to-product ratio
6	Balance_to_Salary	0.076	Wealth distribution indicator
7	Geography_Germany	0.065	Germany has higher churn rate
8	Tenure	0.058	Years with bank
9	CreditScore	0.045	Creditworthiness
10	Gender_Male	0.032	Gender differences

4.3 Exploratory Data Analysis Insights

Geographic Analysis

Churn rates vary significantly by country:

Country	Churn Rate	Insight
Germany	32.4%	Highest churn rate among three countries
France	16.2%	Moderate churn rate
Spain	16.7%	Similar to France

Demographic Patterns

Gender differences in churn behavior:

- Female customers: 25.0% churn rate
 - Male customers: 16.5% churn rate
- Women are approximately 1.5x more likely to churn than men.

Engagement Impact

Customer activity is a strong predictor:

- Inactive members: 26.9% churn rate
 - Active members: 14.3% churn rate
- Active members are nearly 2x less likely to churn.

Product Usage Analysis

Number of products correlates with churn:

- Single-product customers: 27.7% churn rate
 - Multi-product customers: ~12% churn rate
- Customers with 2-3 products show significantly lower churn rates.

Zero Balance Customers

A critical finding: 3,617 customers (36%) have zero account balance.

- Zero-balance customers represent a high-risk segment
- These customers may have already disengaged from the bank

5. Conclusions and Recommendations

5.1 Summary of Findings

- XGBoost is the best performing model with ROC-AUC of 0.8563
- Number of products is the strongest predictor of churn
- Customer activity status is highly predictive (2x difference)
- Germany has the highest churn rate (32.4%)
- 36% of customers have zero balance (high-risk segment)
- Age shows strong positive correlation with churn (0.285)

5.2 Business Recommendations

Recommendation 1: Proactive Retention for High-Risk Customers

- Implement real-time churn risk scoring using XGBoost model
- Target customers with churn probability > 0.5 for retention campaigns
- Prioritize inactive members and single-product customers

Recommendation 2: Product Optimization Strategy

- Launch cross-selling campaigns for single-product customers
- Create product bundles to increase engagement
- Target 2-3 products as the optimal range

Recommendation 3: Geographic-Specific Interventions

- Conduct customer satisfaction surveys in Germany
- Develop Germany-specific retention programs
- Analyze competitive landscape in high-churn regions

Recommendation 4: Zero-Balance Customer Reactivation

- Create special offers for zero-balance customers
- Implement reactivation campaigns with incentives
- Monitor this segment for early churn signals

Appendix: Dashboard Features

The accompanying Streamlit dashboard provides an interactive interface for ongoing churn monitoring. It is structured into five tabs:

Tab	Features
Dashboard	Key metrics, churn distribution, geographic analysis, age analysis
Risk Calculator	Real-time churn probability for individual customers
Feature Analysis	Correlation analysis and feature distributions
Model Performance	Model comparison table and visualization
What-If Simulator	Scenario analysis for retention strategies

Evaluation Metrics Definitions

Metric	Formula	Purpose
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correctness
Precision	$TP / (TP + FP)$	Quality of churn predictions
Recall	$TP / (TP + FN)$	Ability to find actual churners
F1-Score	$2 * (Precision * Recall) / (Precision + Recall)$	Harmonic mean
ROC-AUC	Area under ROC curve	Model discrimination ability

End of Research Paper